

1) Data structure & Justification:

Data structure choice: 2D Array of characters
char doc[MAX_LINES][MAX_LENGTH];

Justification: we choose a fixed array of string because it allows for very fast.

2) Command set

Command:

I <line-number> <text> = Insert a line

D <line-number> = Delete a line

P = Point

Q = quit

Insert Logic

void insert-line (int line, text)

{

shift-all-lines-down-from(line);

copy-text-into-array (line, text);

total-lines++;

}

Delete Logic:

void delete-line (int line)

{

shift-all-lines-up-from(line);

total-lines--;

}

Display logic:

```
void print-doc()
```

```
{
```

```
    for (i=0 to total_lines)
```

```
    {
```

```
        print(i+1, doc[i]);
```

```
    }
```

```
}
```